

An I-f Startup Method with Compensation Loops for PMSM with Smooth Transition

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The back-EMF based sensorless drive of permanent magnet synchronous machines (PMSMs) is attractive for many industrial applications due to its simplicity and good reliability. For applications such as pumps and fans, in which there is no demand for high dynamic performance in the low-speed range, the I-f method for startup is considered as a much simpler solution than many high-frequency signal injection methods. This paper proposes an I-f startup method that enhances the performance by adopting a frequency compensation loop to improve the system stability and a current vector magnitude compensation loop to obtain a smooth transition to back-EMF based sensorless field-oriented-control (FOC). The experimental results verify the effectiveness of the proposed method and its superior performance compared with the existing I-f starting method.

Keywords: I-f startup, PMSM, sensorless drive, variable speed drives, FOC

1. Introduction

Nowadays, Permanent Magnet Synchronous Machines (PMSMs) have been applied in numerous applications, due to their superior performances in terms of high efficiency, compactness and reliability. To reduce the cost of PMSM drive system and improve the reliability, control without rotor position encoder (sensorless control) is desired.

The most widely studied sensorless drive is based on the field-oriented-control (FOC), where accurate rotor position estimation is vital for a successful application. There are basically two types of rotor position estimation methods, i.e. the back-EMF based method and high frequency signal injection method⁽¹⁾. Back-EMF method is suitable for medium to high speed range applications, where the voltage error caused by phase winding resistance voltage drop and inverter nonlinearity can be safely ignored⁽²⁾. Signal injection method detects the rotor saliency and is suitable for position estimation in low speed range for PMSMs. To achieve full speed range sensorless FOC, the two methods need to be combined which complicates the controller design. Many signal injection methods require several properly designed filters and are not easy to implement. Furthermore, since signal injection method requires rotor saliency, it is challenging to be applied on surface-mounted PMSM (SPMSM) where ideally there is no saliency. For many industrial applications such as fans, pumps where there is no high dynamic performance requirement in low speed range, it may be unnecessary to use the signal injection method, since the goal is only to start the machine. Instead, an easy-to-implement method such as I-f control^{(3)–(8)} may be more suitable.

There have been a few works on I-f control as an auxiliary startup method for back-EMF based FOC. Most of the existing methods use a conventional I-f control, where a fixed amplitude current vector is applied with a pre-defined reference frequency ramp to start the machine. Normally, a large current amplitude is applied and a carefully designed frequency ramp is required to successfully start the machine⁽⁵⁾. However, the open-loop speed control is sensitive to disturbance and system stability is always an issue that needs to be considered. In Refs. (9) (10), a frequency compensation loop based on active power is proposed for V/f control of PMSMs to solve the open loop V/f control mid-frequency instability issue. Similarly, the frequency compensation may also be applied for I-f control to improve the system stability, however the detailed stability analysis and convincing improved performance demonstration have not been seen in the existing literatures.

After a sufficient rotor speed is reached under I-f control, the control can be switched to the back-EMF based sensorless FOC. A smooth transition is desired to prevent unwanted high speed and current variations during this transition. In Ref. (8), a first-order-compensator is used to compensate the reference frequency and the current reference amplitude is adjusted according to frequency reference to force the I-f reference frame position to move close to the estimated rotor position before switching to FOC. A successful convergence requires carefully tuned parameters. In Ref. (5), a method to linearly decrease the I-f current vector amplitude is proposed, as the current amplitude decreases, the current vector will move towards the q-axis to provide the required load torque; when the I-f current vector is close enough to the q-axis, switching to FOC is carried out. Similarly, a nonlinear current reduction method is proposed in Ref. (6) to move the I-f current vector close to the q-axis before the switching action. However since all the existing current regulation

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methods are only to reduce the current vector amplitude, the current vector cannot be stably located on the q-axis before switching, which could be an ideal switching condition for achieving a smooth transition.

In this paper, an I-f startup method for PMSMs with two compensation loops is proposed. A frequency compensation loop utilizing active power is used to improve the system stability. A current amplitude compensation loop based on the so-called inner reactive power is applied to make $i_d = 0$ for a smooth transition to FOC. With these two compensation loops, it is shown that faster startup with less speed oscillation and a very smooth transition can be achieved. Detailed analysis regarding stability on the I-f control considering the frequency compensation is provided based on a simplified linearized model.

The rest of the paper is organized as follows: section 2 introduces the overall control scheme and the general principle of I-f startup and switching to FOC. In Sect. 3, the proposed startup method based on I-f control with two compensation loops is analyzed in detail. In Sect. 4, experiments are carried out to verify the performances of the proposed method. Finally, a conclusion is drawn in Sect. 5.

2. Conventional I-f Startup Strategy

Figure 1 shows an overview of back-EMF sensorless control scheme with a conventional I-f controller. First, the two switches (S1 and S2) are connected to terminal 1, and I-f control is applied to start the machine to a sufficient speed. The two switches are then connected to terminal 2, and the control is switched to back-EMF based FOC. The back-EMF based FOC is widely covered in the literature, and is not further discussed in this paper.

The conventional I-f control is shown in Fig. 1. Similar to V/f control, I-f control is a speed open loop control. Instead of controlling a rotating voltage vector, whose amplitude is proportional to the frequency reference (V/f), a constant amplitude current vector is applied to the machine with a frequency ramp. The closed loop current control is realized in a special reference frame with the current vector always on the q^* -axis (Fig. 2). The transformation angle is obtained by integrating the reference frequency.

The I-f startup and switching to FOC procedure contains three stages:

- (1) Rotor position initialization
- (2) I-f startup with a constant current amplitude
- (3) Current amplitude adjustment and switching to FOC

2.1 Rotor Position Initialization Before applying the I-f control, the initial rotor position needs to be known. For applications like fans and pumps where the rotor is allowed to move initially, an easy method of applying a fixed voltage vector to the machine terminal can align the rotor.

2.2 I-f Startup After the rotor position initialization, a fixed amplitude current vector is applied on the q^* -axis of the I-f reference frame as shown in Fig. 2. The I-f frame d^* -axis angle is defined as in Eq. (1), where ω_e is the reference angular frequency. It can be seen from Eq. (1) that initial position of the d^* -axis is set as -90° , so that the q^* -axis is initially aligned with rotor d -axis.

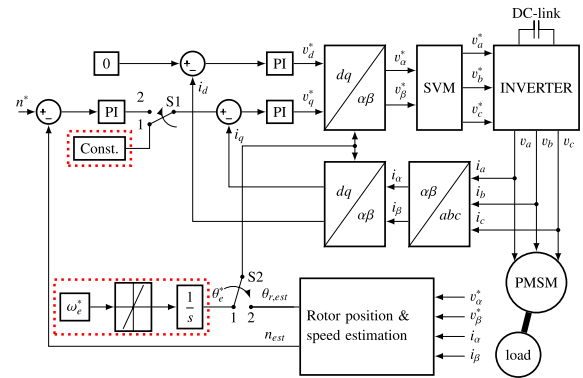


Fig.1. Back-EMF based sensorless control scheme with the conventional I-f startup method

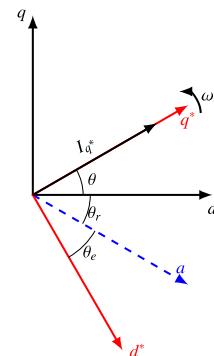


Fig. 2. Reference frame for I-f control

$$\theta_e = \omega_e t - 90^\circ \dots\dots\dots (1)$$

The position difference between the current vector and the rotor d-axis is defined as θ , which can be calculated as in Eq. (2), where θ_r is the rotor position, which is defined as the d-axis position difference with respect to phase-a flux axis as shown in Fig. 2.

$$\theta = \omega_e t - \theta_r \dots \dots \dots (2)$$

The electromagnetic torque generated by the SPMSM under I-f control can be expressed as Eq. (3),

$$T_e = \frac{3}{2} n_{pp} \lambda_m I_{q^*} \sin \theta \dots\dots\dots (3)$$

where n_{pp} is the number of pole pairs; λ_m is the permanent magnet flux magnitude; I_{q^*} is the current vector amplitude on the q^* -axis of the I-f frame.

The basic working principle of I-f control can be illustrated in Fig. 3, where (a) shows after the rotor position initialization, a constant amplitude current vector is applied on the q^* -axis of the I-f frame, since q^* is aligned with the rotor d-axis, no torque will be produced. The integration of reference frequency determines the location of the d^*q^* -axes, the current vector will hence rotate as shown in Fig. 3(b), generating a positive q-axis current. Torque will be generated according to Eq. (3), and when the generated torque is higher than the load torque, the machine will accelerate to follow the reference frequency ramp. It is important to have a large enough current amplitude and that the ramp rate is not too high during the startup, in order to keep the current vector in the first quadrant of the dq frame. I-f control has self-stabilization

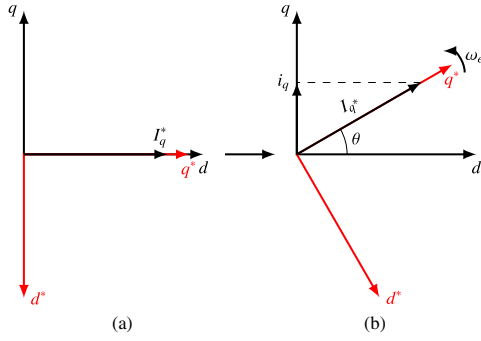


Fig. 3. I-f in operation: (a) I-f control in initial startup (b) I-f control in acceleration

capability, only when the current vector stays in the first quadrant. When there is a positive load torque disturbance, the motor slows down, and the current vector will move closer to the q-axis, generating more torque. A more detailed analysis of the conventional I-f control and the self-stabilization capability can be found in Ref. (5).

Even though the conventional I-f method can start the machine, stability analysis shows that the damping of the system is small, which leads to a slow speed convergence during startup or when load changes, a frequency compensation loop is therefore proposed to enhance the startup performance in this paper, detailed analysis is given in Sect. 3.

2.3 Current Amplitude Regulation and Switching to FOC

After the rotor speed reaches a certain value, the position and speed estimation are normally considered to be accurate. The control can be switched to back-EMF based FOC. But in the case of I-f startup, in steady state, there exists a large d-axis current especially when the machine is lightly loaded. The large d-axis current causes saturation, which may affect the performance of the position estimator. Also, since the desired operation condition for FOC is that the current vector stays on the q-axis, and for I-f control the current vector may stay close to d-axis (e.g., with a light load). If the control is switched directly to FOC when the desired speed is reached, there will be large undesired current transients. Thus it is important to make the current vector close to the q-axis before switching to achieve a smooth transition.

One simple method is to linearly reduce the current amplitude according to Eq. (4) ⁽⁵⁾

$$I_{q^*} = I_0 - K_r(t - t_0) \dots \dots \dots (4)$$

where $t_0 \leq t \leq T_{set} + t_0$, T_{set} is the time interval set for current reduction, I_0 is the initially applied current amplitude and K_r is a constant gain. When the current amplitude is reduced slowly using Eq. (4), because of the self-stabilization mechanism discussed in Sect. 2.2, the current vector has to move closer to the q-axis to produce the required electromagnetic torque. This is illustrated in Fig. 4, where initially the current vector amplitude is I_0 located at position q_0^* , when the current vector amplitude is reduced to I_1 , the vector will be located at position q_1^* . When the current amplitude is further reduced to I_2 , it will move closer to the q-axis and stay at location q_2^* . When the current vector is close enough to the q-axis current (e.g. when it moves to I_2), the control can be switched to FOC. However, this method cannot allocate the current vector on the machine q-axis. This is because it is basically an

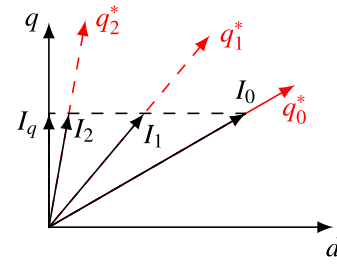


Fig. 4. The current vector moves closer to the q-axis when the current amplitude is reduced under I-f control in steady state

open-loop converging procedure (Eq. (4)) and once the current vector is moved to the second quadrant of the dq-frame, the control will become unstable ⁽⁵⁾. The most ideal switching condition will be that the current vector location is the same before and after the switching. This suggests that preferably, the current vector in the I-f control can be moved to the machine q-axis before switching. This paper proposes a current amplitude compensation method that can ideally fix the current vector on the machine q-axis before switching to FOC, since for SPMSM $i_d = 0$ is the desired working point for FOC. There is no current mismatch before and after switching, the transition can be made much smoother.

3. Proposed I-f Startup Method

The basic control structure of the proposed I-f startup is the same as the conventional one. The frequency reference and current amplitude reference are changed by adding feedback compensations: A frequency compensation loop to improve the system stability, and a current compensation loop for a smooth transition to FOC.

3.1 Frequency Compensation Loop The frequency reference for the proposed I-f control is shown in Fig. 5, as can be seen a frequency compensation (marked with dashed red rectangular) is added to the frequency reference in the conventional I-f method. To understand how the frequency compensation loop works for I-f control, it is helpful to analyze the stability for both the conventional I-f without the frequency compensation loop and the I-f control with the loop.

3.1.1 Stability Analysis of the Conventional I-f Control without the Frequency Compensation Loop The SPMSM model in the synchronous rotor reference frame can be expressed as Eqs. (5)

$$\begin{cases} v_d = R_s i_d + L_d \frac{di_d}{dt} - \omega_r L_q i_q \\ v_q = R_s i_q + L_q \frac{di_q}{dt} + \omega_r (L_d i_d + \lambda_m) \\ T_e = \frac{3}{2} n_{pp} \lambda_m i_q \\ T_e = \frac{J}{n_{pp}} \frac{d\omega_r}{dt} + B_m \frac{\omega_r}{n_{pp}} + T_l \end{cases} \dots \dots \dots (5)$$

where v_d, v_q are d- and q-axes voltages; i_d, i_q are d- and q-axes currents; L_d, L_q are d- and q-axes inductances; R_s is the phase winding resistance; ω_r is the rotor speed; J is the inertia; B_m is the viscous damping coefficient; T_e is the electromagnetic torque generated by the machine, and T_l is the load torque respectively.

Transforming the dq model to the I-f reference frame shown in Fig. 2, the dynamic model of PMSM under I-f frame can be written as Eqs. (6),

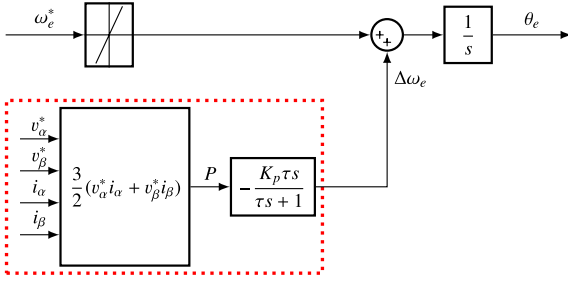


Fig. 5. I-f frequency reference with the frequency compensation loop

$$\begin{cases} \frac{d}{dt}i_{d^*} = -\frac{R_s}{L}i_{d^*} + \omega_e i_{q^*} + \frac{\lambda_m}{L}\omega_r \cos\theta + \frac{v_{d^*}^*}{L} \\ \frac{d}{dt}i_{q^*} = -\omega_e i_{d^*} - \frac{R_s}{L}i_{q^*} - \frac{\lambda_m}{L}\omega_r \sin\theta + \frac{v_{q^*}^*}{L} \\ \frac{d}{dt}\omega_r = A(i_{q^*} \sin\theta - i_{d^*} \cos\theta) - \frac{n_{pp}}{J}T_l - \frac{1}{J}B_m\omega_r \\ \frac{d}{dt}\theta = \omega_e - \omega_r \end{cases} \dots\dots\dots (6)$$

where v_{d^*} , v_{q^*} , i_{d^*} and i_{q^*} are d^* - and q^* -axes voltages and currents respectively; $A = \frac{n_{pp}}{J} \frac{3}{2} \lambda_m$; ω_e is the applied current frequency; L is the dq-axes inductance, $L = L_d = L_q$.

Under I-f control, the d^* - and q^* -axes currents are controlled to be zero and a constant with two PI controllers, respectively. Since the current dynamics are much faster than the mechanical dynamics, the current dynamics can be neglected. Thus the full dynamic model Eqs. (6) can be simplified as Eqs. (7), where $I_{q^*,0}$ is the steady state q^* -axis current.

$$\begin{cases} \frac{d}{dt}\omega_r = A I_{q^*,0} \sin\theta - \frac{n_{pp}}{J}T_l - \frac{1}{J}B_m\omega_r \\ \frac{d}{dt}\theta = \omega_e - \omega_r \end{cases} \dots\dots\dots (7)$$

Linearizing the simplified model in Eqs. (7) around the steady state working point, the linearized model can be written as Eqs. (8),

$$\begin{cases} \frac{d}{dt}\Delta\omega_r = A I_{q^*,0} \cos\theta_0 \Delta\theta - \frac{n_{pp}}{J}\Delta T_l - \frac{1}{J}B_m\Delta\omega_r \\ \frac{d}{dt}\Delta\theta = \Delta\omega_e - \Delta\omega_r \end{cases} \dots\dots\dots (8)$$

where θ_0 is the angle difference between the steady state current vector and the rotor d-axis.

Under conventional I-f control where there is no frequency compensation, i.e. $\Delta\omega_e = 0$, the model in Eqs. (8) can be visualized using a block diagram as shown in Fig. 6, where K is defined as Eq. (9).

$$K = \frac{3}{2} n_{pp} \lambda_m I_{q^*,0} \cos\theta_0 \dots\dots\dots (9)$$

The closed loop transfer function for the linearized model shown in Fig. 6 can be written as Eq. (10).

$$\frac{-\Delta\omega_r(s)}{\Delta T_l(s)} = \frac{\frac{n_{pp}}{J}s}{s^2 + \frac{B_m}{J}s + \frac{Kn_{pp}}{J}} \dots\dots\dots (10)$$

The poles of the transfer function can be calculated as Eq. (11).

$$p_{1,2} = \frac{-\frac{B_m}{J} \pm \sqrt{\left(\frac{B_m}{J}\right)^2 - \frac{4Kn_{pp}}{J}}}{2} \dots\dots\dots (11)$$

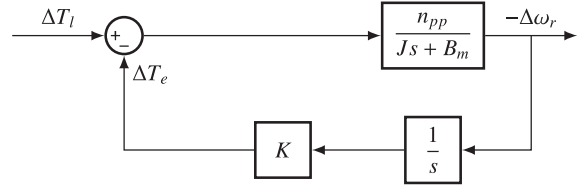


Fig. 6. Simplified linearized model I-f control without the frequency compensation loop

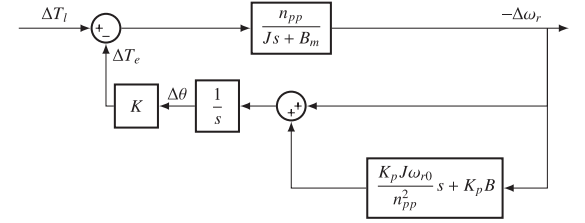


Fig. 7. Simplified linearized model for I-f control with the frequency compensation loop

For the system to be stable, the poles need to have negative real parts, from Eq. (11) it can be seen that the condition is $K > 0$, which means $\theta_0 \in (-90^\circ, 90^\circ)$ according to Eq. (9). One can also notice that even though the system is stable for $\theta_0 \in (-90^\circ, 90^\circ)$, the damping of the system is small since the real part of poles is $-\frac{B_m}{2J}$, which is usually very small. This indicates speed would slowly converge to steady state for conventional I-f control without the frequency compensation loop.

3.1.2 Stability Analysis for I-f Control with the Frequency Compensation Loop To add damping for I-f control the frequency command can be adjusted by using perturbation component of active power as shown in Eq. (12)

$$\Delta\omega_e = -K_p \Delta P \dots\dots\dots (12)$$

The perturbation component of active power ΔP can be calculated as Eq. (13)⁽¹⁰⁾,

$$\Delta P = \frac{J\omega_{r0}}{n_{pp}^2} \frac{d}{dt}\Delta\omega_r + \frac{2B_m\omega_{r0}}{n_{pp}^2} \Delta\omega_r + \frac{T_{l0}}{n_{pp}} \Delta\omega_r \dots\dots (13)$$

where ω_{r0} and T_{l0} are steady state rotor speed and steady state load torque respectively.

By using the active power feedback, the small signal model shown in Fig. 6 becomes Fig. 7, where $B = \frac{2B_m\omega_{r0}}{n_{pp}^2} + \frac{T_{l0}}{n_{pp}}$. The transfer function can be derived as Eq. (14).

$$\frac{-\Delta\omega_r(s)}{\Delta T_l(s)} = \frac{\frac{n_{pp}}{J}s}{s^2 + \left(\frac{\omega_{r0}K_pK}{n_{pp}} + \frac{B_m}{J}\right)s + \frac{n_{pp}K}{J}(K_pB + 1)} \dots\dots (14)$$

Thus the poles can be calculated as Eq. (15), and it can be seen that the poles can be located to the left of $-\frac{B_m}{2J}$ by making the term $\omega_{r0}K_pK$ a constant, achieving faster speed convergence.

$$p_{1,2} = -\frac{\frac{\omega_{r0}K_pK}{n_{pp}} + \frac{B_m}{J}}{2} \pm \frac{\sqrt{\left(\frac{\omega_{r0}K_pK}{n_{pp}} + \frac{B_m}{J}\right)^2 - 4\frac{n_{pp}K}{J}(K_pB + 1)}}{2} \dots\dots (15)$$

In practice, the perturbation component of active power ΔP

is obtained by filtering the instantaneous active power by a first-order high pass filter as shown in Fig. 5. To effectively add damping to the system, ΔP obtained from the high pass filter needs to contain the speed variation information as indicated by Eq. (13). This means the cut-off frequency of the high pass filter needs to be lower than the rotor speed oscillation frequency, which is the system (Eq. (10)) damped natural frequency ω_d expressed by Eq. (16),

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} \dots \dots \dots (16)$$

where $\omega_n = \sqrt{\frac{Kn_{pp}}{J}}$ is the undamped natural frequency and $\zeta = \frac{B_m}{2J\omega_n}$ is the damping ratio of the system, respectively. Once the high pass filter is tuned in such a way, the gain K_p can be tuned to obtain a satisfactory speed damping. The damping ratio ζ_1 of the system with the frequency compensation can be derived as Eq. (17).

$$\zeta_1 = \frac{\frac{\omega_n K_p K}{n_{pp}} + \frac{B_m}{J}}{2 \sqrt{\frac{n_{pp} K}{J} (K_p B + 1)}} \dots \dots \dots (17)$$

3.2 Current Amplitude Compensation and Switching to FOC

After the rotor speed reaches a certain speed, e.g. 20% of rated speed, the control can be switched to FOC. To ensure a smooth transition, the current amplitude is adjusted by using a compensation loop using inner reactive power shown as in Fig. 8, where the compensation loop is marked in the dashed red rectangular.

The inner reactive power is defined as Eq. (18)⁽¹¹⁾. As can be seen from Eq. (18), the inner reactive power is essentially a d-axis current estimator. By controlling the inner reactive power to zero, the d-axis current can be controlled indirectly to zero.

$$\begin{aligned} Q_{inner} &= \frac{3}{2}(v_q i_d - v_d i_q) - \frac{3}{2}\omega_r L_d(i_d^2 + i_q^2) \\ &= \frac{3}{2}\omega_r \lambda_m i_d \dots \dots \dots (18) \end{aligned}$$

Note that the calculation of inner reactive power requires rotor speed. In practice, the reference speed ω_e^* is used, which makes the estimator only valid in steady state. This is not an issue for the startup procedure. The amplitude compensation loop is enabled after the speed reaches steady state. Also, since obtaining the dq-axes voltages and currents require the

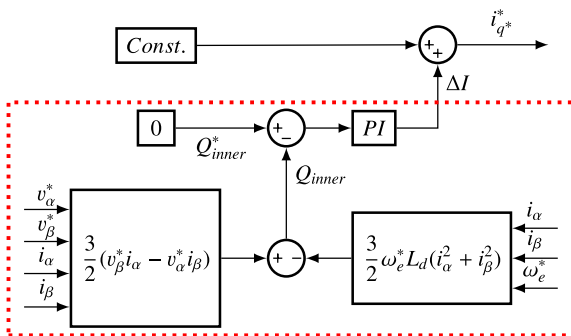


Fig. 8. I-f current amplitude reference with the current amplitude compensation loop

rotor position, Eq. (18) is calculated using $\alpha\beta$ voltages and currents as shown in Fig. 8.

After $i_d = 0$ control is realized under the proposed I-f control, the control can be switched to FOC. As the working condition is $i_d = 0$ before switching, which is the desired working condition for FOC after switching, ideally there is no current or speed mismatch between the two control modes, the transition could be very smooth.

4. Experimental Verification

In this section, experimental results are shown to verify the effectiveness of the proposed method. The improved performance is highlighted by comparing with the existing I-f method. The SPMSM is controlled by a 2-level inverter through dSPACE 1202. The DC-link voltage is $V_{dc} = 550$ V. The switching frequency and sampling frequency is 5 kHz. The dead time is $2 \mu s$ and dead time compensation is not employed in the experiment. The machine parameters are listed in Table 1.

The cut-off frequency of the high pass filter is chosen as 2.5 Hz, i.e. the time constant $\tau = 0.0637$ s. The gain for Eq. (12) is chosen as $K_p = \frac{40}{\omega_e}$ ($\omega_e \neq 0$). The PI controller parameters for the current amplitude compensation loop (Fig. 8) are tuned to be $K_{p,ca} = 0.05$ and $K_{i,ca} = 0.1$ based on principles in Ref. (12).

4.1 I-f Startup Performance (No Switching to FOC)

To verify the effectiveness of the frequency compensation loop, a comparison of I-f startup performance with/without the frequency compensation loop is carried out in the following tests.

The machine rotor is first aligned with the stator phase-a axis by applying a constant voltage vector. Then a current vector of 6 A is applied on the q^* -axis of the I-f reference frame; a 30 Hz frequency reference is then applied with/without the frequency compensation loop. Three different frequency ramp rates, 50 Hz/s, 80 Hz/s and 100 Hz/s are tested.

Figure 9 shows the performance of conventional I-f startup method without the frequency compensation loop. As can be

Table 1. Machine Parameters

SPMSM			
Rated power	2.8 KW	Stator resistance	0.68 Ω
Rated current	7.4 A	d-axis inductance	5.5 mH
Rated voltage	380 V	q-axis inductance	5.5 mH
Rated speed	4500 rpm	PM flux linkage	0.1151 Wb-t
Rated frequency	300 Hz	Inertia	0.01482 kgm^2
Rated torque	5.8 Nm	Pole pairs	4

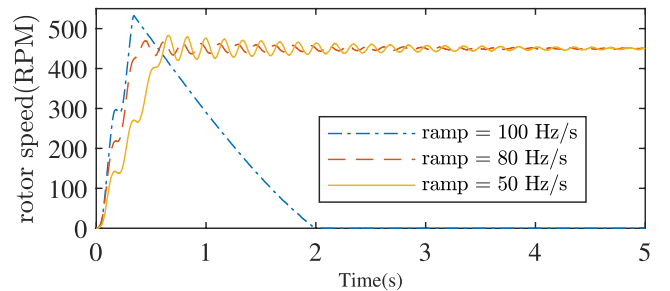


Fig. 9. Conventional open loop I-f startup without the frequency compensation loop

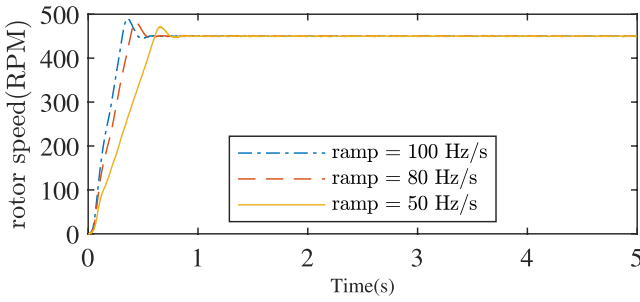


Fig. 10. Proposed I-f startup with the frequency compensation loop

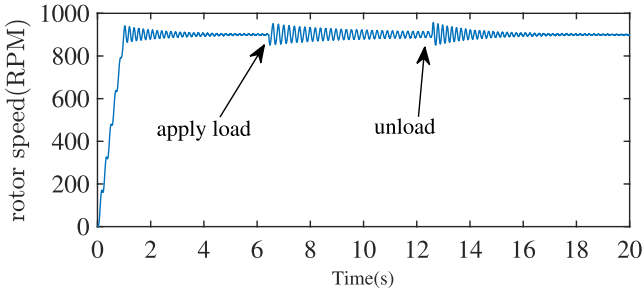


Fig. 11. Conventional open loop I-f startup without the frequency compensation loop with 2 Nm load step disturbance

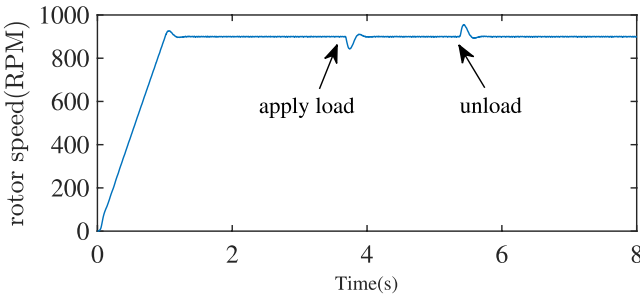


Fig. 12. Proposed I-f startup with the frequency compensation loop with 2 Nm load step disturbance

seen from Fig. 9, the conventional I-f method shows speed oscillation. The speed slowly converges to the reference speed. This slow convergence is expected since the damping is small for conventional I-f control according to the previous stability analysis. Figure 10 shows the results of the proposed I-f startup method with the frequency compensation loop. It can be seen, with the frequency compensation loop, the speed oscillation is greatly reduced, and the rotor speed converges much faster to the reference speed. This agrees well with the stability analysis that the frequency compensation can add damping to the speed in Sect. 3. Also, it is interesting to note that the frequency compensation loop can enable a faster startup, since when the frequency ramp rate is 100 Hz/s, the startup failed for conventional I-f method (Fig. 9), whereas I-f method with the frequency compensation loop succeeded (Fig. 10).

Another test is done for I-f control with/without the frequency compensation loop against load steps. The application target for the paper is fan-, pump-type load, which experiences low torque at low speed. In the experiment, a 2 Nm step load is applied and released to test the performance. Figure 11 and Fig. 12 show that both I-f control methods can

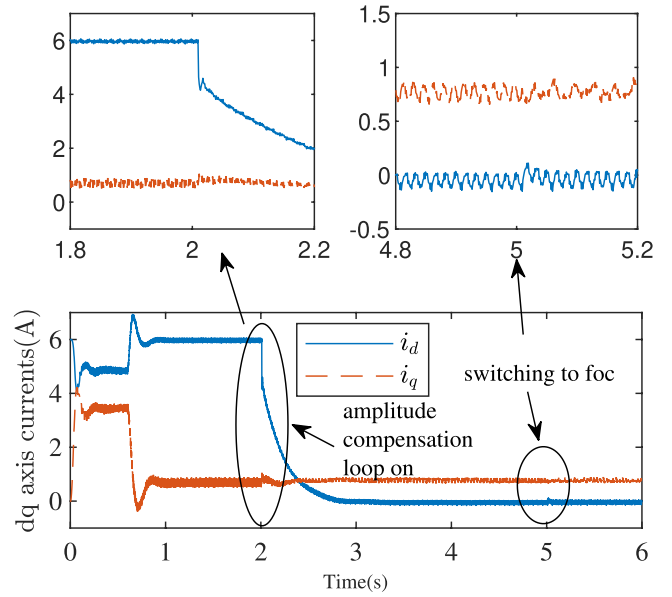


Fig. 13. Machine d- and q-axes currents for the proposed I-f to FOC operation under no load condition

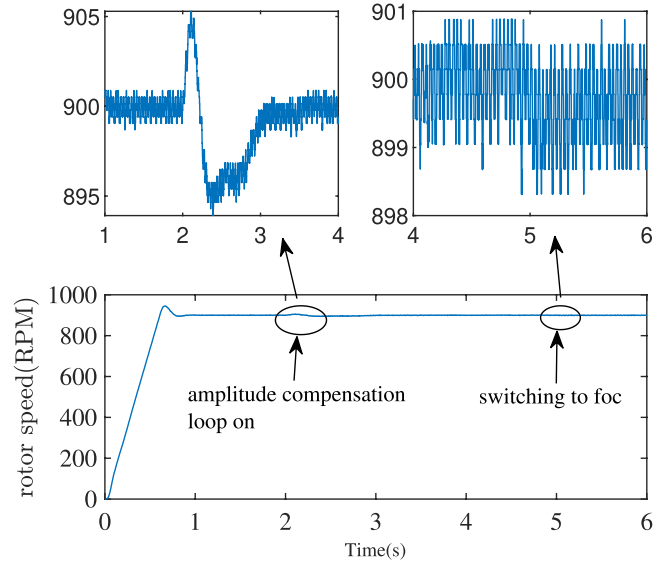


Fig. 14. Rotor speed for the proposed I-f to FOC operation under no load condition

survive 2 Nm load step disturbance, but with the frequency compensation loop, the speed oscillation can be greatly reduced and faster speed convergence can be achieved when load changes.

4.2 Switching Performance Figures 13 and 14 show the dq-axes currents and speed waveforms of the I-f control with the two compensation loops under no load during startup and switching. The machine is accelerated to 900 RPM (20% rated speed) with I-f control with the frequency compensation loop. The ramp rate is set as 100 Hz/s. From Fig. 13 it can be seen, when the current amplitude is enabled at 2 s, the d-axis current reduces to zero in 1 second. At 5 s, the control is switched to sensorless FOC. During switching, the current transients zoomed in Fig. 13 and speed transient shown in Fig. 14 are both very small. This smooth transition is expected, since before switching to FOC, the I-f control is already working in $i_d = 0$ condition.

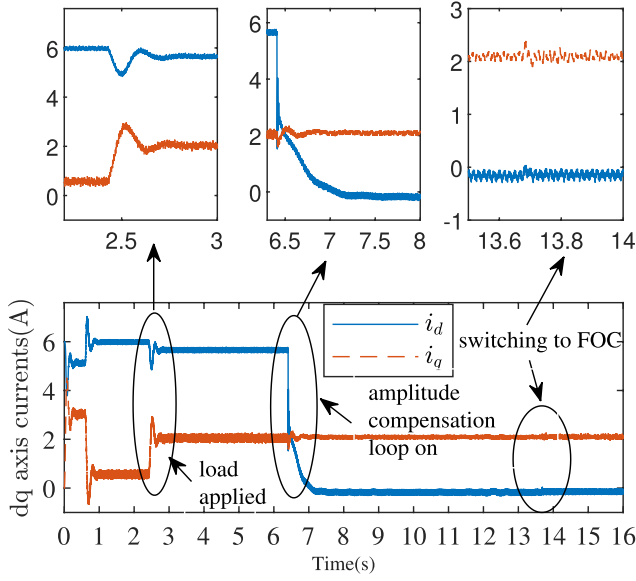


Fig. 15. Machine d- and q-axes currents for the proposed I-f to FOC operation under load condition

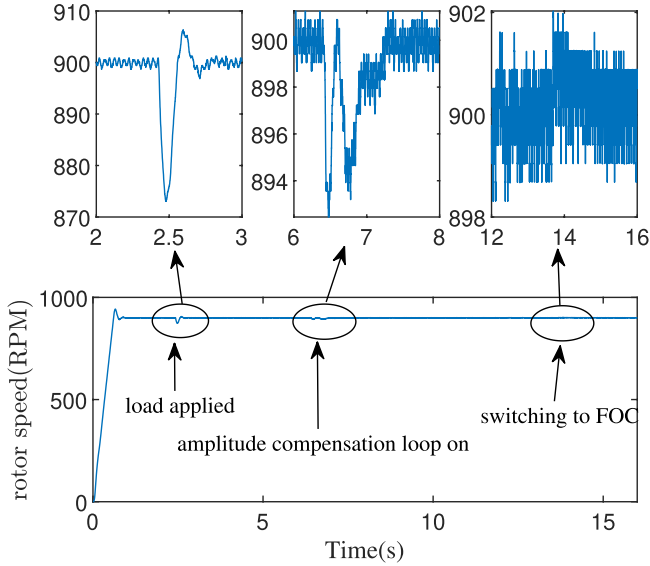


Fig. 16. Rotor speed for the proposed I-f to FOC operation under load condition

Another test is done to show the switching performance under load condition. The machine is accelerated to 900 RPM with I-f control with the frequency compensation loop. Then 2 Nm load is applied at around 2.5 s. When the speed is stable, the current amplitude compensation is enabled to make $i_d = 0$ before switching. Figures 15 and 16 show the dq-axes currents and rotor speed, respectively, for the startup and switching operation. It can be seen at around 6.5 s, when the current amplitude compensation loop is enabled, the d-axis current is reduced to zero, and the machine can run stably in $i_d = 0$ condition. At around 14 s, the control is switched to FOC, the current transient in Fig. 15 and speed transient in Fig. 16 are very small. A very smooth transition is achieved.

To further illustrate the performance, a switching to FOC by linearly decreasing the current amplitude⁽⁵⁾ is done for comparison. The machine is accelerated to 900 RPM with I-f control (with the frequency compensation loop). When

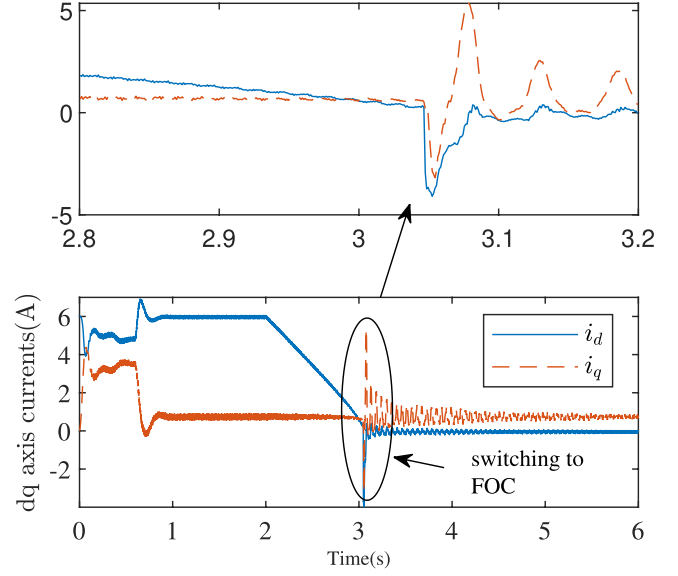


Fig. 17. Machine d- and q-axes currents for I-f to FOC operation using the linear current reduction method before switching

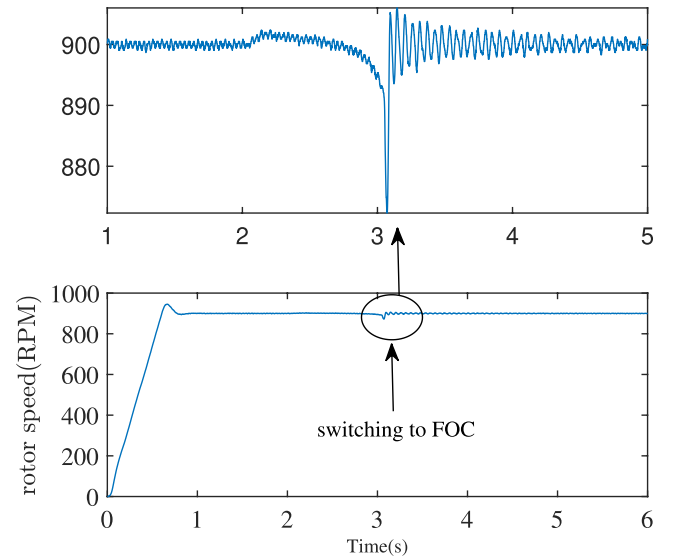


Fig. 18. Rotor speed for I-f to FOC operation using the linear current reduction method before switching

the speed is stable, the current amplitude is reduced using Eq. (4), where the initial current amplitude before current reduction is 6 A, the gain for reduction is $K_r = 5$, and the time interval for current reduction is $T_{set} = 1$ s.

Figures 17 and 18 show the dq-axes currents and speed during the startup and switching, respectively. From Fig. 17, it can be seen when the current amplitude reduction is enabled at 2 s, d-axis current is reduced and q-axis current is intended to be maintained to produce the required torque. However, it can be in Fig. 18 that a speed transient exists and the speed keeps dropping as d-axis current is decreasing, as explained in Ref. (5). The switching to FOC is done at 3 s, when the d-axis current is about 0.3 A. It can be seen that there are still quite some large current and speed transients when switching to FOC. This is because even though $i_d = 0.3$ is close to $i_d = 0$, there is still some mismatch between the currents

between the I-f and FOC modes. More importantly, the speed mismatch will result in larger changes in q-axis reference current. Comparing with this conventional linear current reduction method in Ref. (5), the proposed current amplitude compensation method can make sure that the machine runs stably with $i_d = 0$ before switching to FOC, thus a much smoother transition can be realized.

5. Conclusion

A startup method based on I-f control with two compensation loops is proposed in this paper. The frequency compensation loop improves the dynamic and stability performance in the startup period. A detailed stability analysis based on a simplified I-f control model is provided. The current amplitude compensation loop based on inner reactive power is employed to make $i_d = 0$ for I-f control to ensure a very smooth switching. The proposed method is easy to implement and could be a good candidate for fan, pump applications.

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